

# Quick Check

## Lesson 5: Primary and Secondary Growth: Why Does Maize Thicken but Not Make Wood?

*7 multiple-choice questions • choose A, B, C or D*

Sub-Strand 2.2: Growth and Development in Plants | Lesson 5

**On Mr. Wafula's farm, a maize seedling grows taller each week. Which part of the plant is responsible for making the shoot longer?**

- A** The vascular cambium in the stem
- B** The pith in the centre of the stem
- C** The bark around the stem
- D** The apical meristem at the shoot tip

**A student looks at a stem section under a microscope. The vascular bundles are scattered through the stem and there is no ring of cambium. Which stem is most likely being viewed?**

**A** Older Rosecoco bean stem with rings

**B** Grevillea robusta (silky oak)

**C** Cypress

**D** Maize H614

**In a stem section of *Grevillea robusta*, where is the vascular cambium found?**

- A** Outside the bark
- B** Between the xylem and the phloem
- C** In the centre of the pith
- D** Inside the epidermis only, away from the vascular tissues

**Each year, the vascular cambium of a Grevillea tree divides. What does it add, and what is the result?**

- A** Secondary xylem outward and bark inward, making roots longer
- B** Only phloem, so the stem stretches without making wood
- C** Secondary xylem inward and secondary phloem outward, increasing girth and forming annual rings
- D** Primary xylem and primary phloem, making the tree taller

**Maize H614 has no vascular cambium and makes no wood, yet its stem widened from 6 mm to 22 mm. How can this be explained?**

- A** Maize makes annual rings inside the stem
- B** The stem only looked wider because it was stretched
- C** The cambium is hidden and makes wood slowly
- D** The stem thickened by primary growth, through cell division and enlargement behind the apical meristem

The width of a maize H614 stem was 6 mm on Day 14, 14 mm on Day 28 and 22 mm on Day 42. By how many millimetres did the stem widen between Day 14 and Day 42?

A 16 mm

B 8 mm

C 22 mm

D 28 mm

**A group is revising its Day 1 model of how Rosecoco bean grows. Its stem was 2 mm, 4 mm and 6 mm wide on Days 14, 28 and 42. Which statement should it add?**

- A** The bean stem is thin because it has no vascular bundles
- B** The bean stem has scattered bundles and no cambium, like maize
- C** The bean stem thickens a lot because its cambium is as active as in a Grevillea tree
- D** The bean stem has a cambium, but it is only slightly active at the base, so thickening is small

# All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).